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Table 9.2: Types and characteristics of tissue systems				
S.No.	**Types/ Characters**	**Epidermal tissue system**	**Ground or fundamental tissue system**	**Vascular or conduction tissue system**
1.	Formation	Forms the outermost covering protoderm	Forms the ground meristem	Forms the procambial bundles
2.	Components	epidermal cells, stomata and epidermal outgrowths	Simple permanent tissues – Parenchyma and Collenchyma	Xylem and Phloem
3.	Functions	Protection of plant body; absorption of water in roots; gas exchange for photosynthesis and respiration; transpiration in shoots	Gives mechanical support to the organs; prepares and stores food in leaf and stem	Conducts water and food; gives mechanical strength

9.4 Epidermal Tissue System

Introduction

Epidermal tissue system is the outer most covering of plants. It is in direct contact with external environment. It consists of epidermis derived from protoderm. Epidermis is derived from two Greek words, namely 'Epi' and 'Derma'. 'Epi' means *upon* and 'Derma' means *skin*. Although epidermis is a continuous outer layer, it is interrupted by stomata in many plants.

Root Epidermis

The outer layer of the root is known as ***piliferous layer* or *epiblema.*** It is made up of single layer of parenchyma cells which are arranged compactly without intercellular spaces. It is devoid of epidermal pores and cuticle. Root hair is always single celled, it absorbs water and mineral salts from the soil. The another important function of *piliferous layer* is *protection.*

Stem Epidermis

It is protective in function and forms the outermost layer of the stem. It is a single layer of parenchymatous rectangular cells. The cells are compactly arranged without intercellular cells. The outer walls of epidermal cells have a layer called *cuticle*. The cuticle checks *transpiration*. The cuticle is made up of *cutin*. In many plants it is also mixed wax to form epicuticular wax. Epidermal pores may be present here and there. Epidermal cells are living. Chloroplasts are usually absent except in guard cells of stomata. In many plants a large number of epidermal hairs occur on the epidermis.

Leaf Epidermis

The leaf is generally *dorsiventral*. It has upper and lower epidermis. The epidermis is usually made up of a single layer of cells that are closely packed. Generally the